

"Basic Electrical Engineering"

①

* MODULE-1

→ D.C. CIRCUITS:

An electric circuit is an interconnection of the various elements such as, voltage source, current source, resistors, inductors & capacitors.

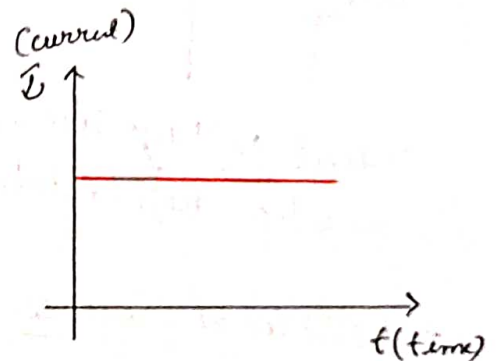
* In an electric circuit, two types of currents may flow

- i) Direct current (D.C)
- ii) Alternating current (A.C).

* Accordingly, there are two types of circuits, (i) D.C circuits
ii) A.C circuits.

* The flow of D.C characterizes the flow of electric charge in one particular direction.

* A direct current always remains constant & does not vary with time. So frequency of D.C is zero.



* Elements of an Electric circuit:

→ An Electric circuit consists of two types of elements

- (i) Active Elements @ Source
- (ii) Passive Elements @ Sink.

→ i) Active Elements are the elements of the circuit which produces energy of their own & can impart it to other elements.

Ex: voltage source & current source.

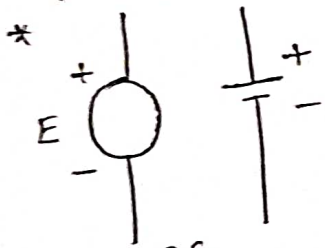
→ ii) Passive elements of an electric circuit does not produce energy of their own. They consume energy from source.

Ex: Resistor, inductor & capacitor.

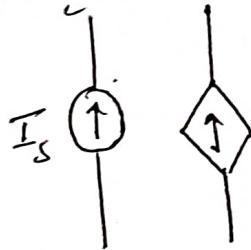
* Ideal Source:

* Ideal voltage source: The terminal voltage is constant, irrespective of the current delivered by it to load.
* The internal resistance is zero in ideal voltage source. ($r=0$)

→ Ideal current source: it delivers a constant current to the circuit, irrespective of the load connected to its terminals.
* The internal resistance is ∞ (type) in ideal current source ($r=\infty$)



Ideal ^{DC} voltage source
Symbolic Representation



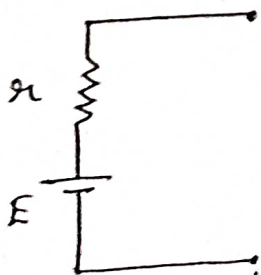
Ideal DC current source
Symbolic Representation.

* Practical Source:

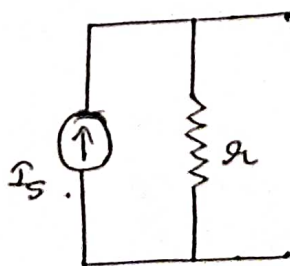
→ Practical source have internal resistance ' r '.

→ The internal resistance of a "voltage source" is always connected in series with it.

→ The internal resistance of a "current source" is always connected in "parallel" with it



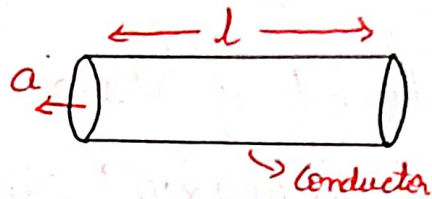
Practical voltage source.



Practical current source

Resistance (R): Resistance is the property of a conductor by virtue of which, it opposes the flow of current through it.
→ The unit of resistance is "ohm (Ω)".

→ The Resistance of a conductor is directly proportional to its length (l) & inversely proportional to its area of cross section.



$$\therefore R \propto \frac{l}{a} \quad \text{or} \quad R = \rho \frac{l}{a}$$

Where ρ is a constant, known as the "specific resistance" or "resistivity" of the material of the conductor.

→ Unit of resistivity is "ohm-meter" ($\Omega\text{-m}$)

* Resistivity (ρ):

Consider a equation $R = \rho \frac{l}{a} \rightarrow \text{①}$

In equation ① if $l = 1\text{m}$ & $a = 1\text{m}^2$, then $\rho = R$.

Hence, the resistivity of a conductor may be defined as its resistance, when its length is unity & area of cross section is also unity.

* Electric current (I):

The rate at which electrical charge is transferred across a point in a conductor is known as the current.

$$i = \frac{dq}{dt} \quad \text{or} \quad I = \frac{q}{t}$$

The unit of current is "Ampere".

* One Ampere: One ampere of current, is defined as the current flowing through a conductor, when a charge of one ~~not~~ coulomb crosses across a point in conductor in one second.

$$1 \text{ A} = \frac{1 \text{ C}}{1 \text{ Sec}}$$

* 1 coulomb of charge is equal to charge of 6.242×10^{18} electrons.

* $1 \text{ e} = -1.602 \times 10^{-19} \text{ Coulomb}$.

* Electric Potential: It always refers to a point in a charged conductor.

→ Electric potential is defined as the "work done" to bring a unit positive charge from infinity to that point.

→ The unit of electric potential is "volt".

* Potential Difference (V): The potential difference b/w any two points of a charged conductor is the "amount of work that has to be done to bring a unit positive charge from the point of lower potential to the point of higher potential."

→ The unit of potential difference is "volt".

→ ^{Electric} Potential $\therefore \dots \dots (V) = \frac{\text{work done}}{\text{charge}} = \frac{W}{Q}$

$$\therefore \boxed{V = \frac{W}{Q}}$$

→ $1 \text{ volt} = 1 \text{ joule} / 1 \text{ coulomb}$.

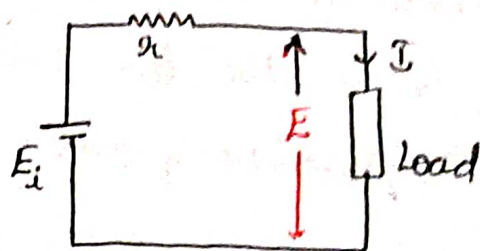
* E.M.F (Electromotive Force) of source (E)

→ E.M.F of the source is the voltage available across its terminals.

→ The voltage available across the terminals of a voltage source ⁽³⁾ is slightly less than its internal voltage E_i , due to the small voltage drop across its internal resistance.

→ The unit of EMF is also "volt".

$$\rightarrow \boxed{E = E_i - I r}$$



* POWER (P): Power is defined as the rate at which, the work is done.

$$E = V = \text{voltage}$$

$$\boxed{P = E I = \frac{E^2}{R} = I^2 R}$$

→ Its unit is watt (W)

* Energy (W): Energy is the capacity to do work.
→ It is equal to the total work done in a particular time.

$$\boxed{W = P t = E I t = \frac{E^2}{R} t = I^2 R t}$$

→ Unit of Energy is "kilo watt hour (kWh)".

$$\rightarrow 1 \text{ unit} = 1 \text{ kWh} = 3.6 \times 10^6 \text{ Joules}$$

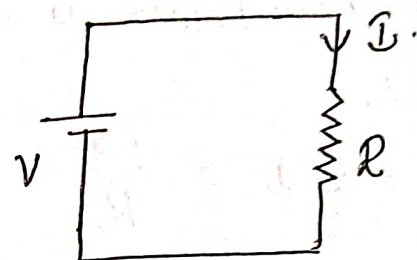
* OHM'S LAW

* Statement: It states that the current flowing through any conductor is directly proportional to the potential difference between the two ends of the conductor, provided its temperature & other physical parameters remains constant.

Take a simple D.C circuit as shown in the figure, we can say that:

$$I \propto V \rightarrow \textcircled{1}$$

$$\text{i.e. } I = \left(\frac{1}{R}\right) V$$



D.C circuit

Where R is a constant, known as Resistance of the conductor.

$$\text{i.e. } \boxed{I = \frac{V}{R}}$$

$$\textcircled{2} \quad \boxed{V = IR}$$

→ Ohm's Law can be applied for both A.C & D.C circuits.

* Limitations of Ohm's Law:

- 1) Ohm's Law does not hold good for non-linear devices such as semiconductors (diodes), Zener diodes & voltage regulators etc.
- 2) It is not applicable for non-metallic conductors, such as silicon carbide.
- 3) It does not hold good to arc lamps.
- 4) It does not hold good where the temperature rise is rapid in some metals.

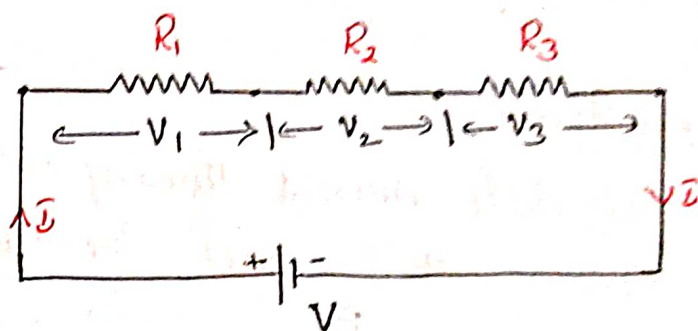
Series circuit: The circuit in which resistors are connected end-to-end, so that there is only one path for current flow, is called "series circuit".

* In Series circuit:

- 1) Same current flows through all the resistances.
- 2) The voltage drop across each resistance is different.
- 3) The sum of the voltage drop is equal to applied voltage.

Consider a series circuit.

Where resistors R_1 , R_2 & R_3 are connected in series & voltage V is applied across the combination.



I is the current flowing through them.

Let V_1 , V_2 & V_3 be the voltage drop across resistors R_1 , R_2 & R_3 respectively.

$$\therefore \text{Now } V = V_1 + V_2 + V_3 \quad \longrightarrow (1)$$

$$\text{W.K.T, According to Ohm's Law } V = IR \quad \longrightarrow (2)$$

$$V = IR_1 + IR_2 + IR_3$$

$$\therefore V = I(R_1 + R_2 + R_3)$$

$$\frac{V}{I} = R_1 + R_2 + R_3$$

According to Ohm's Law, $\frac{V}{I}$ is the total Resistance of the circuit

$$\therefore \boxed{R_T = R_1 + R_2 + R_3}$$

i.e. total Resistance = Sum of individual resistance.

* If there are n resistances connected in series, then total Resistance is given by

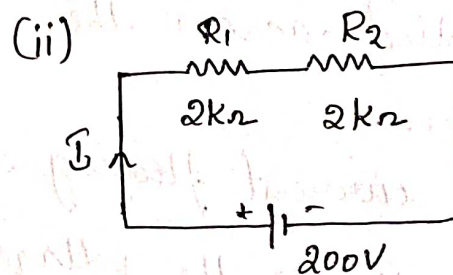
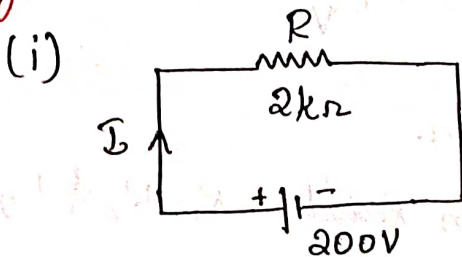
$$\therefore R_T = R_1 + R_2 + R_3 + \dots + R_n$$

* If there are " n equal resistances" connected in series, then total Resistance is given by.

$$R_T = nR.$$

* Problems:

1) calculate current flowing in each of the circuit, shown in figure given that each resistor is of $2k\Omega$.

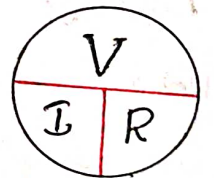


Solution:

$$(i) \quad I = \frac{V}{R} = \frac{200}{2 \times 10^3} = 100 \text{ mA}$$

$$(ii) \quad \text{Total Resistance } (R) = R_1 + R_2 = 2k\Omega + 2k\Omega = 4k\Omega.$$

$$\therefore I = \frac{V}{R} = \frac{200}{4 \times 10^3} = 50 \text{ mA}$$



Ohm's circle

calculate the voltage across each of the resistors shown in fig & then calculate the supply voltage V .

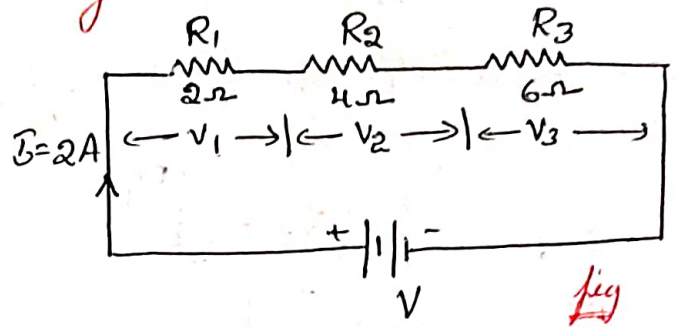
Soln: According ohm's law

$$V_1 = IR_1 = 2 \times 2 = 4V$$

$$V_2 = IR_2 = 2 \times 4 = 8V$$

$$V_3 = IR_3 = 2 \times 6 = 12V$$

$$\therefore \text{Total supply voltage } (V) = V_1 + V_2 + V_3 = 4 + 8 + 12 = 24V$$



3) For the circuit shown in fig find the circuit current (I), given that the supply voltage 200V.

Soln: Total resistance of circuit is

$$R_T = R_1 + R_2 + R_3 = 40 + 60 + 80$$

$$\therefore R_T = 180$$

According ohm's law $I = \frac{V}{R_T} = \frac{200}{180} = 1.11A$

$$\therefore I = 1.11A$$

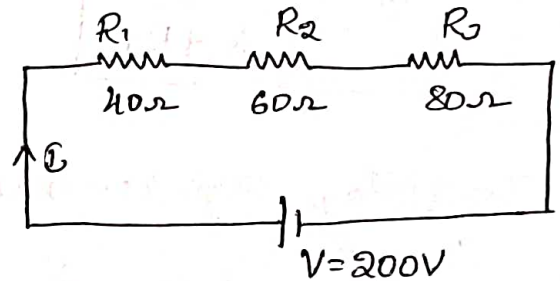
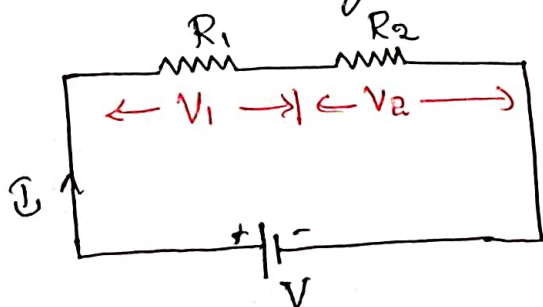


Fig:

→ Voltage Division between Two Resistors:

→ Let us consider a circuit as shown in fig, which deals with the division of voltage between only two resistors R_1 & R_2 connected in series across the supply voltage V .

→ We need to find voltage drop across the Resistor R_1 .



→ The total Resistance of the circuit is

$$R = R_1 + R_2 \quad \text{--- (1)}$$

→ Hence, the current in the circuit is

$$I = \frac{V}{R} = \frac{V}{R_1 + R_2} \quad \text{--- (2)}$$

→ The voltage drop across the resistor R_1 is given by

$$V_1 = I R_1 \rightarrow (3)$$

∴ Substitute eqn (2) in eqn (3), we get

$$V_1 = \left[\frac{V}{R_1 + R_2} \right] R_1$$

$$\therefore \boxed{V_1 = \frac{V R_1}{R_1 + R_2}}$$

114 ^{drop} V_2 across Resistor R_2 is given by.

$$\boxed{V_2 = \frac{V R_2}{R_1 + R_2}}$$

∴ The Voltage drop across Resistor $R_1 [V_1] = \frac{\text{Main voltage} \times \text{Resistance of the same Branch}}{\text{Sum of the Resistance}}$

* Problem:

① In the voltage divider of fig shown below, given that $V = 50V$ & V_2 across R_2 is $20V$. calculate the value R_1 .

Soln: $V = 50V$, $V_2 = 20V$, $R_2 = 100\Omega$, $R_1 = ?$

Apply voltage division Rule across Resistance R_2

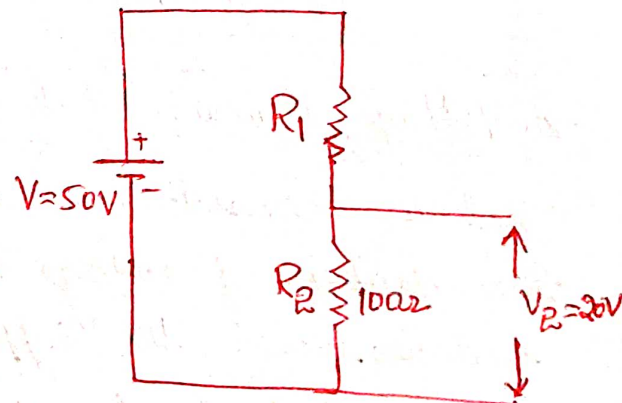
$$\therefore V_2 = \frac{V \times R_2}{R_1 + R_2}$$

$$20 = \frac{50 \times 100}{R_1 + 100}$$

$$\therefore R_1 + 100 = \frac{50 \times 100}{20} \Rightarrow R_1 + 100 = \frac{5000}{20}$$

$$\therefore R_1 + 100 = 250$$

$$\therefore R_1 = 250 - 100 \Rightarrow \boxed{R_1 = 150\Omega}$$



Parallel Circuit

6

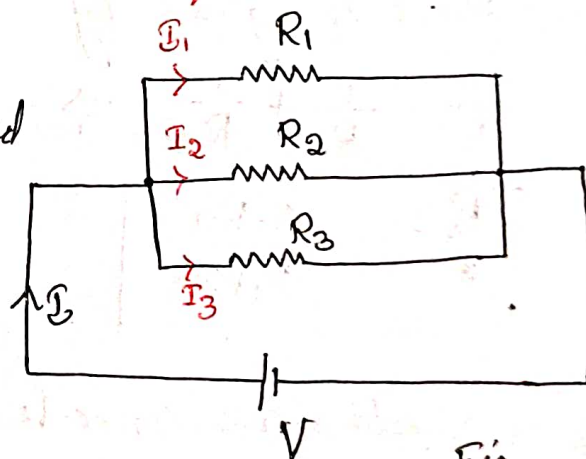
→ When number of resistors are connected in such a way that one end of each resistor is joined to common point & other end of each resistor is joined to ^{another} common point, then resistors are said to be connected in parallel & such circuits are known as "Parallel circuit".

→ In parallel circuit, the total current is divided into the number of paths equal to the number of resistances in parallel.

→ The total current is always sum of all the individual current. i.e. $I = I_1 + I_2 + I_3 + \dots + I_n$.

* Expression for an equivalent resistance of parallel circuit:

→ Consider a circuit as shown in fig, let resistors R_1, R_2 & R_3 are connected in parallel across a voltage source of V volts.



Fig

→ The total current flowing in circuit is I Amperes.

→ Let I_1, I_2 & I_3 are the current in resistors R_1, R_2 & R_3 respectively.

$$\therefore \text{Now total current } (I) = I_1 + I_2 + I_3 \quad \longrightarrow \textcircled{1}$$

N.K.T According to Ohm's law

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2} \quad \& \quad I_3 = \frac{V}{R_3}$$

$$\therefore \text{eqn } \textcircled{1} \text{ becomes } I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$I = V \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]$$

$$\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad \longrightarrow \textcircled{2}$$

As per ohm's law $\frac{V}{I}$ is total resistance R .

so that $\frac{I}{V} = \frac{1}{R}$.

hence eqn (2), becomes.

$$\boxed{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \longrightarrow (3)$$

Hence, when a number of resistors are connected in parallel, the reciprocal of the total resistance is equal to the sum of the reciprocal of the individual resistance.

NOTE:

If there are only two resistance connected in parallel, then total resistance is given by.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R} = \frac{R_2 + R_1}{R_1 R_2}$$

$$\boxed{R = \frac{R_1 R_2}{R_1 + R_2}}$$

If 3 resistors are connected in parallel, then total resistance is given by.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \Rightarrow \frac{1}{R} = \frac{R_2 R_3 + R_1 R_3 + R_1 R_2}{R_1 R_2 R_3}$$

$$\boxed{R = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}}$$

current Distribution in Parallel circuits

(7)

Let us consider a circuit, in which two resistors R_1 & R_2 be connected in Parallel across a voltage source of V volts.

→ I is current flowing through circuit.

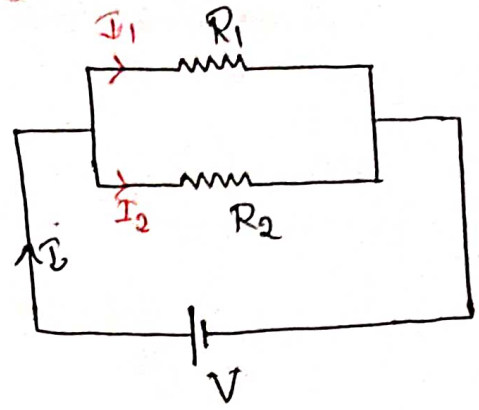


Fig:

V.K.T in Parallel circuit

$$V = V_1 = V_2 \longrightarrow (1)$$

where $V_1 \rightarrow$ voltage drop across R_1 & $V_2 \rightarrow$ voltage drop across R_2 .

Total Resistance of the circuit is given by.

$$R_T = \frac{R_1 R_2}{R_1 + R_2} \longrightarrow (2)$$

$$\therefore \text{As per Ohm's Law } V = I R_T \longrightarrow (3)$$

Substitute eqn (2) in eqn (3)

$$\therefore V = I \left[\frac{R_1 R_2}{R_1 + R_2} \right] \longrightarrow (4)$$

Consider voltage drop across Resistance R_1

$$\text{i.e. } V_1 = I_1 R_1 \longrightarrow (5)$$

Equating eqn (4) & (5) we get $\therefore V = V_1$

$$\therefore V = V_1$$

$$I \left[\frac{R_1 R_2}{R_1 + R_2} \right] = I_1 R_1 \Rightarrow$$

$$\therefore \boxed{I_1 = \frac{I R_2}{R_1 + R_2}}$$

Similarly

$$\boxed{I_2 = \frac{I R_1}{R_1 + R_2}}$$

IMP
 $\therefore$ current through Resistance $R_1 (I_1) = \frac{\text{Main current} \times \text{Resistance of other Branch}}{\text{Sum of the Resistance}}$

Problems:

① Find the supply current I & branch current I_1 & I_2 in circuit as shown in fig.

Soln: In parallel circuit, v.k.T

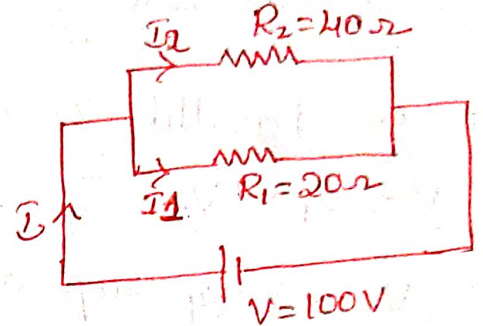
$$I = I_1 + I_2$$

$$V = V_1 = V_2 = 100V$$

$$\therefore I_1 = \frac{V_1}{R_1} = \frac{100}{20} = 5A$$

$$I_2 = \frac{V_2}{R_2} = \frac{100}{40} = 2.5A$$

$$\text{Supply current } I = I_1 + I_2 = 5 + 2.5 = 7.5A$$



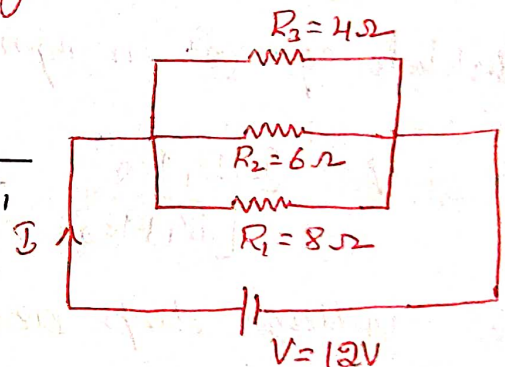
② Find the effective resistance & the supply current to the circuit as shown in fig.

Soln Effective Resistance (R) = $\frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$

$$\therefore R = \frac{8 \times 6 \times 4}{8 \times 6 + 6 \times 4 + 4 \times 8}$$

$$R = 1.85\Omega$$

$$\therefore \text{Supply current } (I) = \frac{V}{R} = \frac{12}{1.85} = 6.49A$$



③ A current of 10A is divided b/w two resistors in the circuit. find the current in each of the two resistors.

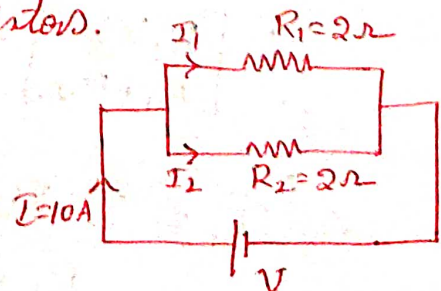
Soln: In parallel ckt, $I = I_1 + I_2$.

Apply current division Rule.

$$I_1 = \frac{I R_2}{R_1 + R_2} = \frac{10 \times 2}{2 + 2} = 5A$$

$$I_2 = I - I_1 = 10 - 5 = 5A$$

$\therefore$ Equal resistances share the supply current equally.



(2) A current of 10A is divided b/w two resistors in the ckt shown in fig. find the current in each of the two resistors.

Soln: $I = I_1 + I_2$

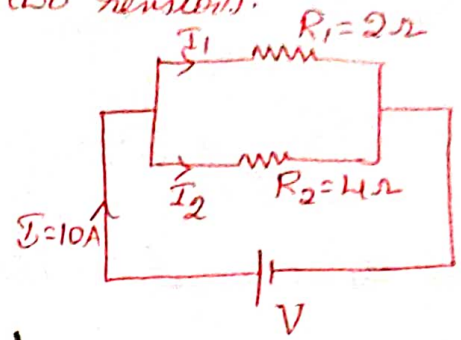
Apply current division Rule

$$\therefore I_1 = \frac{I \times R_2}{R_1 + R_2} = \frac{10 \times 4}{2 + 4} = 6.66 \text{ A}$$

$$I_2 = \frac{I \times R_1}{R_1 + R_2} = \frac{10 \times 2}{2 + 4} = 3.34 \text{ A}$$

$$\textcircled{00} \quad I_2 = I - I_1 = 10 - 6.66$$

$$I_2 = 3.34 \text{ A}$$



$\therefore$ Lesser the resistance, the greater its share of the supply current.

* Series & Parallel circuit:

→ It may be often necessary to use combinations of series & parallel arrangement of resistors in circuits.

→ Such circuit can be solved by combining the parallel group first into an equivalent resistance & then adding this equivalent resistance to other series resistance to determine the total Resistance.

→ Then current produced by the applied voltage can be obtained as usual.

Problems: Problems:

① Find the resistance of the circuit shown i.e. R_{AB} .

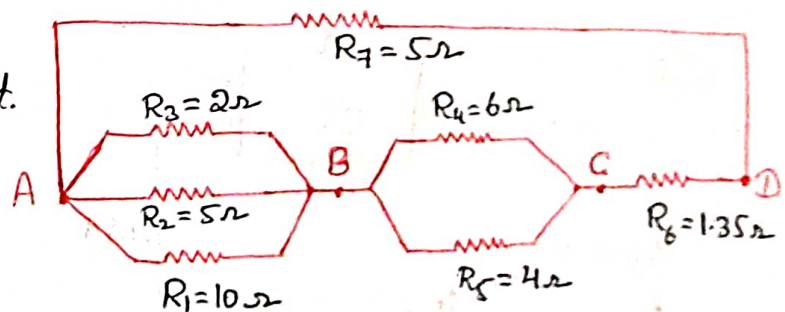
Soln:

→ Find resistance b/w terminal A & B first.
i.e. R_{AB} .

i.e. R_1, R_2 & R_3 are connected in parallel

$$\therefore R_{AB} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1} = \frac{10 \times 5 \times 2}{10 \times 5 + 5 \times 2 + 2 \times 10} = \frac{100}{50 + 10 + 20} = \frac{100}{80}$$

$$R_{AB} = 1.25 \Omega$$

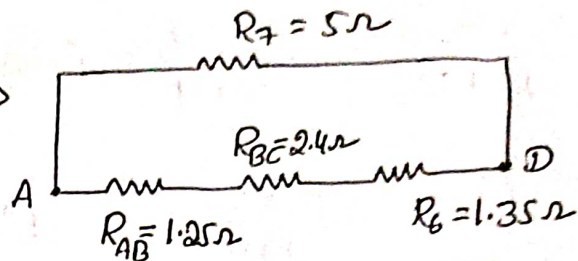


→ find resistance b/w terminal B & D. i.e R_{BC}

$$R_4 \parallel R_5 \quad \therefore R_{BC} = \frac{R_4 R_5}{R_4 + R_5} = \frac{6 \times 4}{6 + 4} = \frac{24}{10}$$

$$R_{BC} = 2.4 \Omega$$

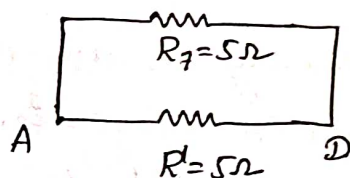
hence the circuit becomes $\Rightarrow$



Now R_{AB} , R_{BC} & R_6 are connected in series,

$$\text{hence } R' = R_{AB} + R_{BC} + R_6 = 1.25 + 2.4 + 1.35 = 5 \Omega$$

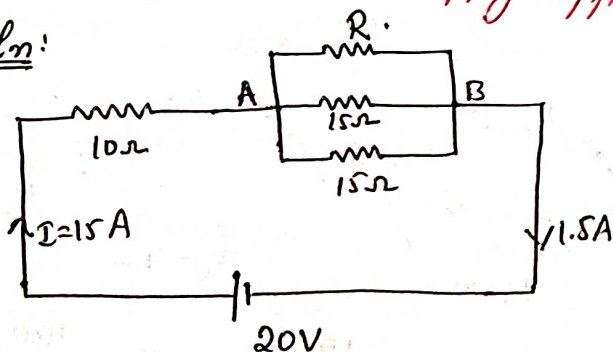
hence circuit becomes $\Rightarrow$



$$\therefore \text{Resistance b/w A \& D is } R_{AD} = \frac{5 \times 5}{5 + 5} = \frac{25}{10} = 2.5 \Omega$$

2) A resistance of 10Ω is connected in series with the two resistors each of 15Ω arranged in parallel, that resistance must be shunted across this parallel combination, so that the total current taken will be $1.5 A$ from $20 V$ supply applied?

Soln:

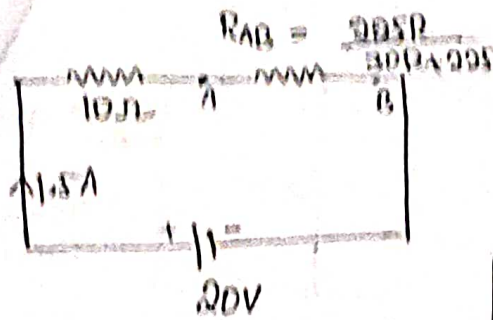


Find equivalent resistance b/w point A & B.

$$R_{AB} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$\therefore R_{AB} = \frac{(R)(15)(15)}{R15 + (15 \times 15) + (15 \times R)} = \frac{225R}{30R + 225}$$

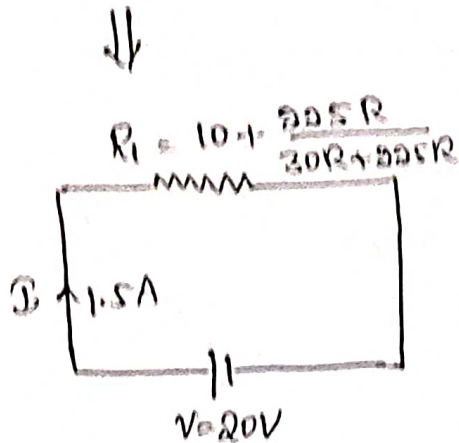
So above circuit can be written as.



∴ The total Resistance of the circuit is

$$R_t = 10 + R_{AB} = 10 + \frac{225R}{30R + 225R}$$

$$R_t = 10 + \frac{225R}{30R + 225R}$$



$$\therefore R_t = \frac{V}{I}$$

$$10 + \frac{225R}{30R + 225} = \frac{20}{1.5}$$

$$\frac{225R}{30R + 225} = 13.33 - 10 \Rightarrow \frac{225R}{30R + 225} = 3.33$$

$$\therefore 225R = 100R + 749.25$$

$$125R = 749.25 \Rightarrow R = \frac{749.25}{125} \Rightarrow \boxed{R = 6\Omega}$$

3) Find the value of Resistor R, so that current drawn from the source is 250mA.

Soln: In given circuit

→ R & 40Ω are Parallel

$$\therefore R \parallel 40 \Rightarrow \frac{40 \times R}{40 + R}$$

→ Now 40Ω & $\frac{40R}{40+R}$ are connected in series in fig: 2

$$\therefore R \cdot 40 + \frac{40R}{40+R} = R_t$$

$$\frac{40(40+R) + 40R}{40+R} = R_t \Rightarrow \frac{1600 + 80R}{40+R} = R_t$$

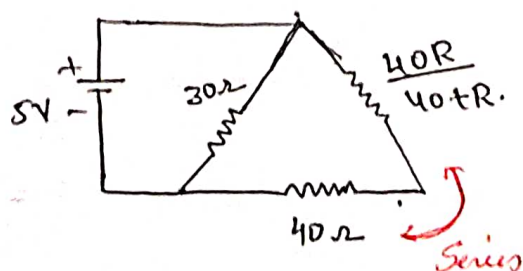
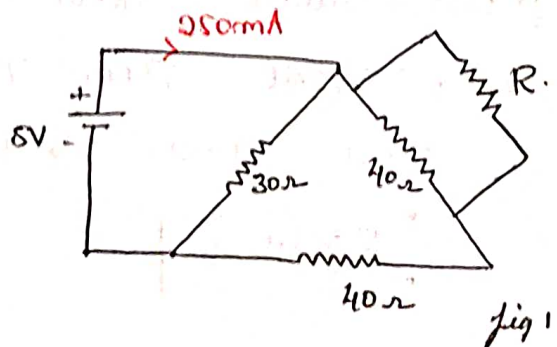
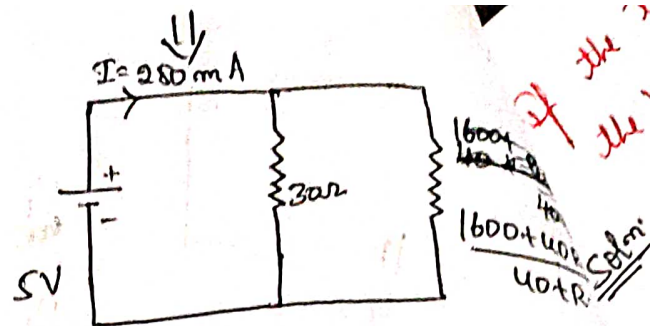


fig 2

$$\text{now } 30 \Omega \parallel \frac{1600 + 80R}{40 + R} \Rightarrow R_{eq} = \frac{30 \times R_t}{30 + R_t}$$

$$\therefore R_{eq} = \frac{30 \times \left(\frac{1600 + 80R}{40 + R} \right)}{30 + \frac{1600 + 80R}{40 + R}}$$



$$R_{eq} = \frac{\frac{48000 + 2400R}{(40 + R)}}{\frac{1200 + 30R + 1600 + 80R}{(40 + R)}} = \frac{48000 + 2400R}{2800 + 110R} = R_{eq}$$

$$R_{eq} = \frac{V}{I} \Rightarrow \frac{48000 + 2400R}{2800 + 110R} = \frac{5}{250 \times 10^{-3}} = 20$$

$$\therefore \frac{48000 + 2400R}{2800 + 110R} = 20 \Rightarrow 48000 + 2400R = 56000 + 2200R$$

$$2400R - 2200R = 56000 - 48000$$

$$200R = 8000 \Rightarrow R = \frac{8000}{200}$$

$$200R = 8000 \Rightarrow R = \frac{8000}{200}$$

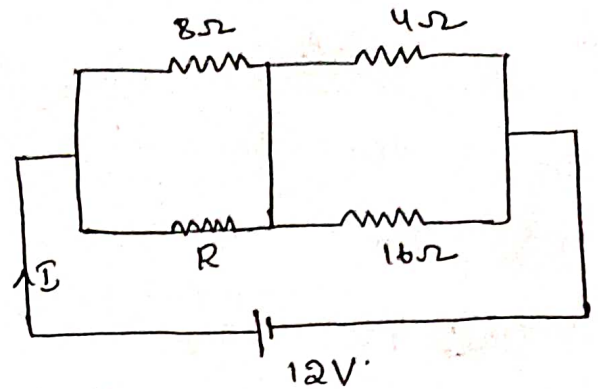
$$200R = 8000 \Rightarrow R = \frac{8000}{200}$$

$$R = 40 \Omega$$

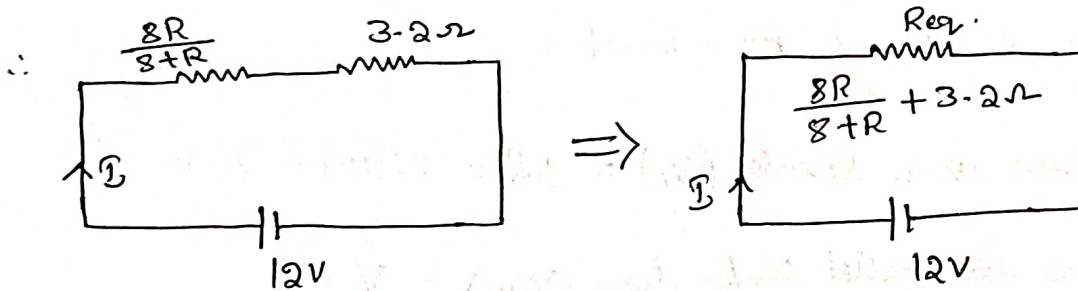
Q. If the total power dissipated in the circuit shown is 18 Watts. find the value of R & its current.

Soln: In given circuit

8Ω & $R\Omega$ are in Parallel and
 4Ω & 16Ω are in parallel.



$$\therefore 8\Omega \parallel R\Omega = \frac{8R}{8+R} \quad \& \quad 4\Omega \parallel 16\Omega = \frac{4 \times 16}{4+16} = 3.2\Omega$$



$$\therefore R_{eq} = \frac{8R}{8+R} + 3.2 = \frac{8R + 25.6 + 3.2R}{8+R} = \frac{25.6 + 11.2R}{8+R}$$

Total power dissipated in ckt i.e $P = 18W$

W.K.T $P = VI$

$$\therefore 18 = 12 \times I$$

$$I = \frac{18}{12}$$

$$\boxed{I = 1.5A}$$

W.K.T $P = \frac{V^2}{R_{eq}}$

$$\therefore R_{eq} = \frac{V^2}{P} = \frac{(12)^2}{18}$$

$$\boxed{R_{eq} = 8\Omega}$$

hence $R_{eq} = \frac{25.6 + 11.2R}{8+R}$

$$8 = \frac{25.6 + 11.2R}{8+R} \Rightarrow 64 + 8R = 25.6 + 11.2R$$

$$\Rightarrow 11.2R - 8R = 64 - 25.6 \Rightarrow 3.2R = 38.4 \Rightarrow R = \frac{38.4}{3.2}$$

$$\boxed{R = 12\Omega}$$

5) Determine the value of R , if the power dissipated in 10Ω resistor is $40W$.

Soln: given data:

$$\rightarrow P_{10\Omega} = 40W$$

$$I^2 R = 40$$

$$I^2 (10) = 40$$

$$I^2 = \frac{40}{10} = 4$$

$$\therefore \boxed{I = 2A} \text{ current through } 10\Omega \text{ Resistor.}$$

$$\rightarrow \text{voltage drop across } 10\Omega \text{ Resistor } (V_1) = IR = (2)(10) = 20V$$

$$\rightarrow \text{voltage drop across the Parallel combination } (V_{AB}) = V - V_1$$

$$= 50 - 20 = 30V$$

$$\therefore \boxed{V_{AB} = 30V}$$

$$\rightarrow \text{current through } 80\Omega \text{ Resistor, } I_1 = \frac{V_{AB}}{80} = \frac{30}{80} = 0.375A$$

$$\text{current through } 40\Omega \text{ Resistor, } I_2 = \frac{V_{AB}}{40} = \frac{30}{40} = 0.75A$$

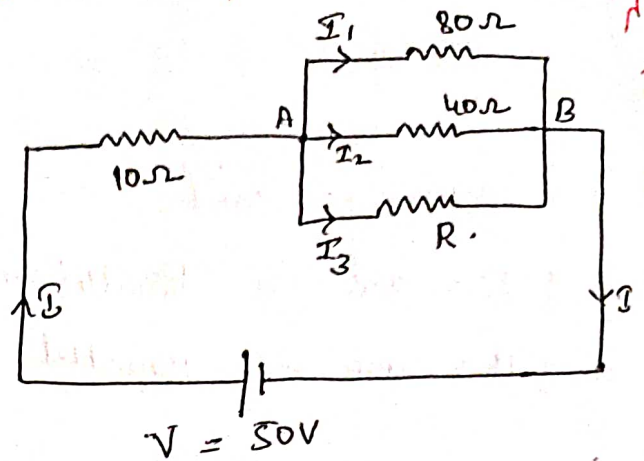
$$\text{W.K.T } I = I_1 + I_2 + I_3$$

$$\therefore I_3 = I - (I_1 + I_2) = 2 - (0.375 + 0.75)$$

$$\boxed{I_3 = 0.875A}$$

$$\therefore \text{The value Resistor } R = \frac{V_{AB}}{I_3} = \frac{30}{0.875} = 34.3\Omega$$

$$\therefore \boxed{R = 34.3\Omega}$$



In the circuit shown, find the value of Resistance R , when the power consumed by 12Ω Resistor is $36W$.

Soln:

given data:

$$\rightarrow P_{12\Omega} = 36W$$

$$\therefore \text{W.K.T } P = \frac{V^2}{R}$$

$$\text{i.e. } P_{12\Omega} = \frac{(V_{AB})^2}{12} \Rightarrow 36 = \frac{(V_{AB})^2}{12} \Rightarrow (V_{AB})^2 = 432$$

$$\therefore \boxed{V_{AB} = 20.78V}$$

$$\rightarrow I_1 = \frac{V_{AB}}{12} = \frac{20.78}{12} = 1.73A$$

$$I_2 = \frac{V_{AB}}{8} = \frac{20.78}{8} = 1.15A$$

$$I_3 = \frac{V_{AB}}{36} = \frac{20.78}{36} = 0.57A$$

$$\therefore \text{Total current in ckt, } (I) = I_1 + I_2 + I_3 = 1.73 + 1.15 + 0.57$$

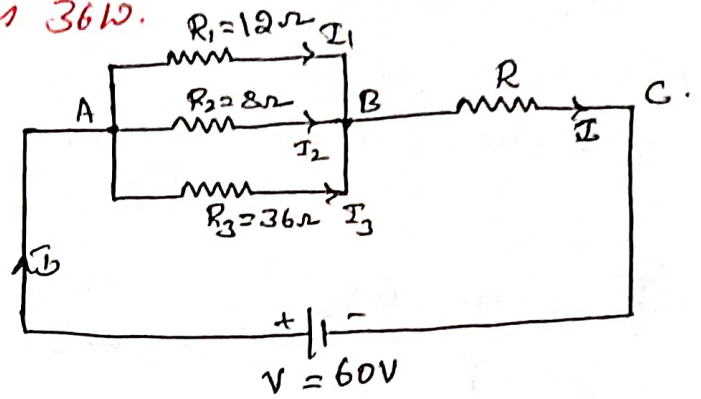
$$\therefore \boxed{I = 3.45A}$$

$$\rightarrow \text{Voltage drop across Resistor, } R \text{ is } V_{BC} = V - V_{AB} = 60 - 20.78$$

$$\boxed{V_{BC} = 39.22V}$$

$$\rightarrow \text{The value of } R \text{ is } = \frac{V_{BC}}{I} = \frac{39.22}{3.45} = 11.36\Omega$$

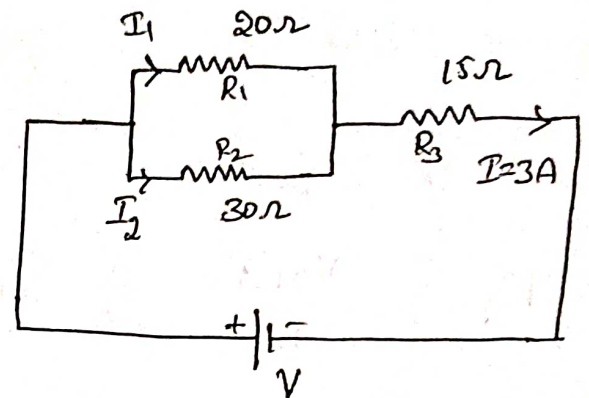
$$\therefore \boxed{R = 11.36\Omega}$$



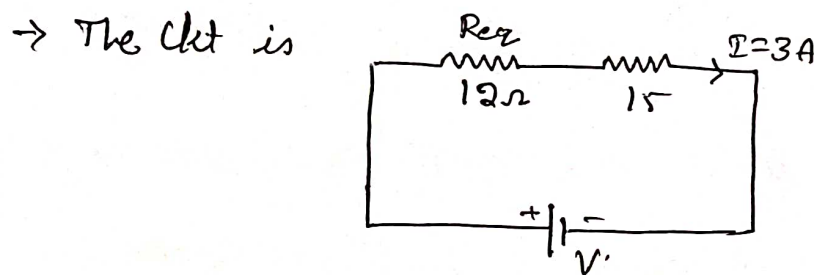
- 7) A circuit consists of two parallel resistors having resistance of 20Ω & 30Ω respectively connected in series with 15Ω . If current through 15Ω resistor is $3A$, find: (i) current in 20Ω & 30Ω resistors. (ii) The voltage across the whole circuit (iii) The total power & power consumed in all resistors.

soln: given data:

$$I = 3A$$



$$\rightarrow 20\Omega \parallel 30\Omega = R_{eq} = \frac{20 \times 30}{20 + 30} = 12\Omega$$



$$\therefore \text{Total Resistance of ckt, } R_t = 12 + 15 = 27\Omega$$

$$\therefore V = I R_t = 3 \times 27 = 81V \quad \therefore \boxed{V = 81V}$$

$$\rightarrow \text{The current through Resistor } 20\Omega, I_1 = \frac{I \times R_2}{R_1 + R_2} = \frac{3 \times 30}{20 + 30} = 1.8A$$

$$\therefore \boxed{I_1 = 1.8A}$$

$$\rightarrow \text{The current through Resistor, } 30\Omega, I_2 = \frac{I \times R_1}{R_1 + R_2} = \frac{3 \times 20}{20 + 30} = 1.2A$$

$$\boxed{I_2 = 1.2A}$$

$$\rightarrow \text{Total power consumed by circuit, } P = VI = 81 \times 3 = 243W$$

$$\boxed{P = 243W}$$

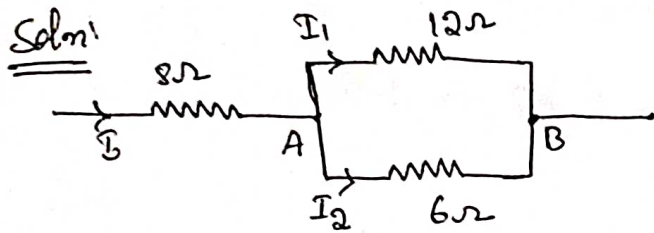
$$\rightarrow P_{30\Omega} = I_2^2 \times 30 = (1.2)^2 \times 30 = 43.2W$$

$$P_{20\Omega} = I_1^2 \times 20 = (1.8)^2 \times 20 = 64.8W$$

$$P_{15\Omega} = I^2 \times 15 = (3)^2 \times 15 = 135W$$

$$P = P_{30\Omega} + P_{20\Omega} + P_{15\Omega} = 43.2 + 64.8 + 135 = 243W$$

A 8Ω resistor is in series with a Parallel combination of two resistors 12Ω & 6Ω . If the current in the 6Ω resistor is $5A$, determine the total power dissipated in the ckt.



given data:

$$I_2 = 5A.$$

voltage across Resistor 6Ω , $V_{AB} = I_2 \times 6$.

$$\therefore V_{AB} = 5 \times 6 = 30V.$$

$$\rightarrow \text{current through Resistor, } 12\Omega \text{ is } I_1 = \frac{V_{AB}}{12} = \frac{30}{12} = 2.5A.$$

$$\rightarrow I = I_1 + I_2 = 2.5 + 5 = 7.5A$$

$$P_{8\Omega} = (I)^2 \times 8 = (7.5)^2 \times 8 = 450W.$$

$$P_{12\Omega} = (I_1)^2 \times 12 = (2.5)^2 \times 12 = 75W.$$

$$P_{6\Omega} = (I_2)^2 \times 6 = (5)^2 \times 6 = 150W.$$

$$\therefore \text{Total power dissipated in ckt. } P = P_{8\Omega} + P_{12\Omega} + P_{6\Omega}.$$

$$P = 450 + 75 + 150 = 675W.$$

$$\therefore \boxed{P = 675W}$$

9) A circuit consists of two parallel resistors having resistance of 20Ω & 30Ω respectively. Connected in series with a 15Ω resistor. If the current through 30Ω resistor is $1.2A$. find

(i) currents in 20Ω & 15Ω resistors

(ii) The voltage across the whole ckt

(iii) voltage across 15Ω resistor & 20Ω resistor.

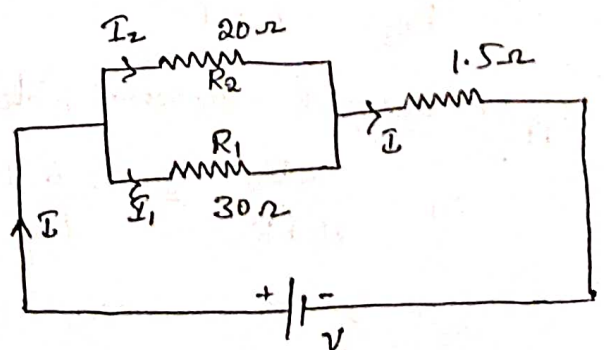
(iv) Total power consumed in the ckt.

Soln: given data: $I_1 = 1.2A$

$\rightarrow V$ across Resistor R_1 is

$$V_1 = I_1 R_1 = (1.2)(30) = 36V.$$

(i) V across parallel resistors is same.



$$\therefore V_2 = V_1$$

$$I_2 R_2 = 36V$$

$$I_2 = \frac{36}{20} = 1.8A$$

$$\therefore \boxed{I_2 = 1.8A}$$

current through 15Ω Resistor is $I = I_1 + I_2 = 1.2 + 1.8$

$$\therefore \boxed{I_{15\Omega} = 3A}$$

(ii) V_g across the whole ckt

$V = V_g$ drop across parallel ckt + V_g drop across 15Ω Resistor

$$V = 36 + (15 \times 3) = 36 + 45$$

$$\boxed{V = 81V}$$

$$\text{iii) } V_g \text{ across } 15\Omega = 15 \times 3 = 45V$$

$$V_g \text{ across } 20\Omega = 20 \times 1.8 = 36V$$

$$\text{iv) Power consumed in the circuit} = I_1^2 \times 30 + I_2^2 \times 20 + I^2 \times 15$$

$$P = (1.2)^2 \times 30 + (1.8)^2 \times 20 + (3)^2 \times 15$$

$$\boxed{P = 243W}$$

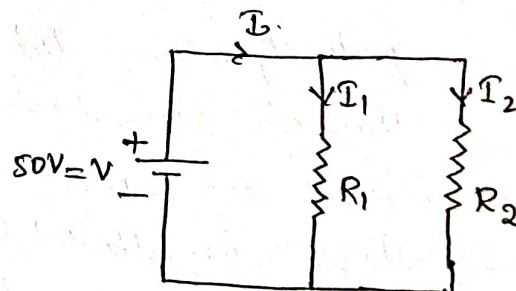
10) Find the total current, current through R_1 & R_2 if $R_1 = 10\Omega$, $R_2 = 20\Omega$ & $V = 50V$.

Soln: $R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 20}{10 + 20}$

$$\therefore \boxed{R_{eq} = 6.67\Omega}$$

$$\therefore I = \frac{V}{R_{eq}} = \frac{50}{6.67} = 7.5A$$

$$\therefore \boxed{I = 7.5A}$$



Apply current division Rule for ckt.

$$I_1 = \frac{I \times R_2}{R_1 + R_2} = \frac{7.5 \times 20}{10 + 20} = 5A$$

$$I_2 = \frac{I \times R_1}{R_1 + R_2} = \frac{7.5 \times 10}{10 + 20} = 2.5A$$

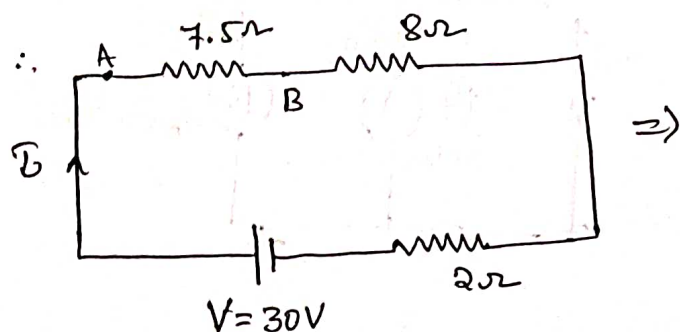
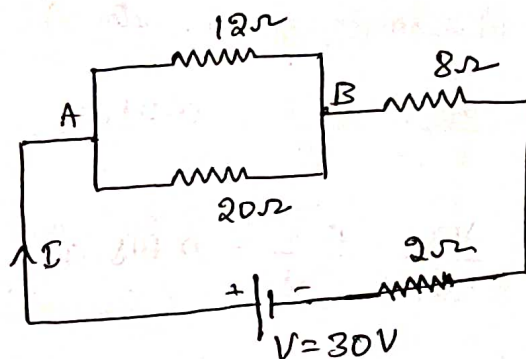
13
A coil of 12Ω resistance is in parallel with a coil of 20Ω resistance. This combination is connected in series with a third coil of 8Ω resistance. The whole circuit is connected across a battery having an emf of $30V$ & internal resistance of 2Ω . Calculate
(a) terminal voltage of the battery (b) Power in 12Ω coil.

Soln:

(a) terminal voltage of the battery

$$R_{AB} = \frac{R_A R_B}{R_A + R_B} = \frac{12 \times 20}{12 + 20}$$

$$R_{AB} = 7.5\Omega$$



$$\text{Now } R_t = 7.5 + 8 + 2 = 17.5\Omega$$

$$R_t = 17.5\Omega$$

$$I = \frac{V}{R_t} = \frac{30}{17.5} = 1.714A$$

$$I = 1.714A$$

→ Internal Resistance of Battery $= 2\Omega = R$.

∴ Internal voltage drop of Battery $= R \times I = 2 \times 1.714 = 3.428V$

∴ Terminal V_t $(V_t) = V - IR = 30 - 3.428$

$$V_t = 26.572V$$

(b) Power in 12Ω coil:

→ current through 12Ω Resistor is, $I_{12\Omega} = \frac{I \times R_B}{R_A + R_B} = \frac{1.714 \times 20}{12 + 20} = 1.07A$

$$I_{12\Omega} = 1.07A$$

→ $P_{12\Omega} = (I_{12\Omega})^2 R_A = (1.07)^2 \times 12 = 13.7W$

$$P_{12\Omega} = 13.7W$$

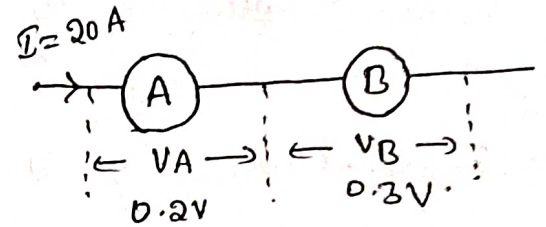
27 A current of 20A flows through two ammeters A & B in series. The potential difference across A is 0.2V & across B is 0.3V. Find how the same current will divide w/o A & B when they are in Parallel.

Soln:

R_A = Resistance of Ammeter A

$$R_A = \frac{V_A}{I} = \frac{0.2}{20} = 0.01 \Omega$$

$$R_B = \frac{V_B}{I} = \frac{0.3}{20} = 0.015 \Omega$$



(a) Ammeters in Series.

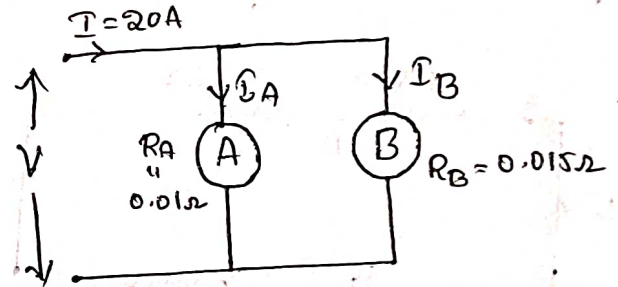
→ Now Ammeters are Connected in Parallel as shown in fig

$$I_A = \frac{I \times R_B}{R_A + R_B} = \frac{20 \times 0.015}{0.01 + 0.015}$$

$$I_A = 12 \text{ A}$$

$$I_B = \frac{I \times R_A}{R_A + R_B} = \frac{20 \times 0.01}{0.01 + 0.015}$$

$$I_B = 8 \text{ A}$$



KIRCHOFF'S LAW:

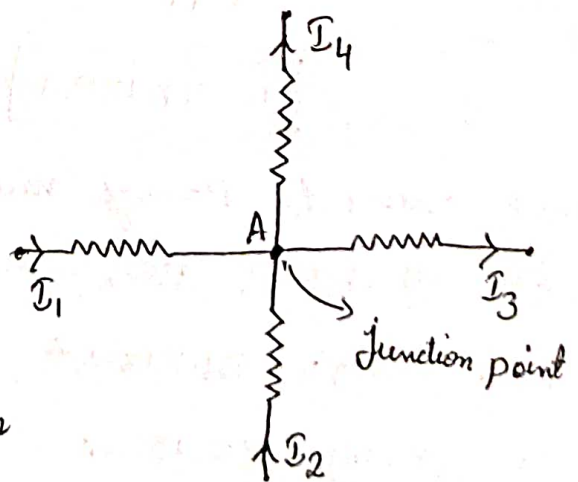
- "Gustav Robert Kirchhoff", a German physicist state two laws which enables us to find the currents flowing in an electric circuit & voltage across the various elements of the circuit.
- Kirchhoff's laws are very useful in solving circuits which cannot be easily solved by using Ohm's Law.
- There are two Kirchhoff's laws 1) Kirchhoff's current law (KCL)
2) Kirchhoff's Voltage Law (KVL).

1) Kirchhoff's Current Law (KCL):

→ Statement: "At any junction of an electric circuit, the sum of all the currents entering the junction is equal to the sum of all the currents leaving the junction".

→ Kirchhoff's current law can also be stated as "The algebraic sum of all the current meeting at any junction of an electrical circuit is zero".

→ Fig shows the junction A of an electric circuit at which four currents I_1, I_2, I_3 & I_4 meet.



→ All current flowing towards junction point A are taken +ve & all current flowing away from junction are taken as -ve.

→ Then according to KCL.

$$I_1 + I_2 - I_3 - I_4 = 0$$

i.e.

$$\boxed{\sum I = 0}$$

$$\textcircled{28} \quad I_1 + I_2 = I_3 + I_4 =$$

Problem
1) Find the currents in the various branches of the given network as shown in fig.

Soln → Apply KCL at various nodes.

Let the current distribution is as shown in fig(2)

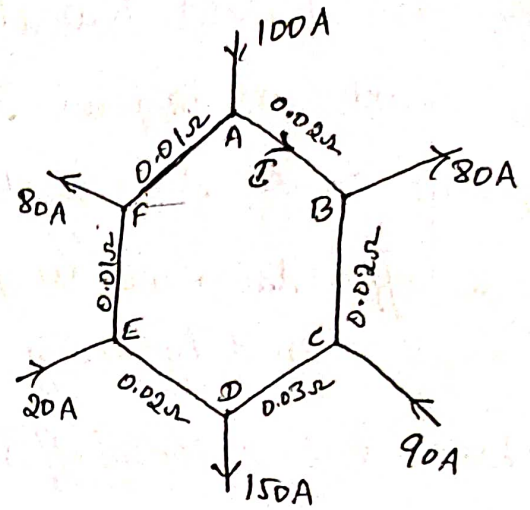
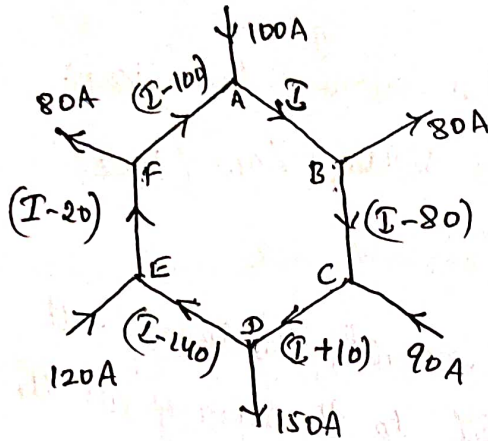


Fig: (1)

→ Apply KVL to loop ABCDEFA.

$$0.02(I) + 0.02(I - 80) + 0.03(I + 10) + 0.02(I - 140) + 0.01(I - 20) + 0.01(I - 100) = 0$$

$$0.11I - 5.3 = 0$$

$$I = \frac{5.3}{0.11} = 48.1818 \text{ A}$$

$$\therefore \boxed{I = 48.1818 \text{ A}}$$

→ Hence current through various branches are,

$$AB = 48.1818 \text{ A}, \quad BC = -31.8182$$

$$BC = (I - 80) = -31.8182 \text{ A}$$

$$CD = (I + 10) = 58.1818 \text{ A}$$

$$DE = (I - 140) = -91.8182 \text{ A}$$

$$EF = (I - 20) = 28.1818 \text{ A}$$

$$FA = (I - 100) = -51.8182 \text{ A}$$

-ve sign indicates that the direction is opposite to that assumed in fig(2).

* Kirchoff's Voltage Law (KVL):

→ Statement: In any closed electrical circuit, the algebraic sum of voltage drops plus the algebraic sum of all the emf's in that circuit is zero.

i.e. Algebraic sum of emf's + Algebraic sum of voltage drops = 0

$$\boxed{\sum E + \sum IR = 0}$$

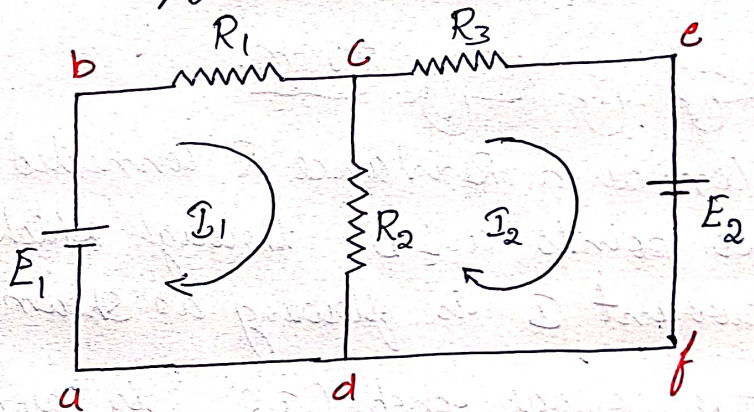
(08)

In any closed electrical circuit, the algebraic sum of voltage drop across circuit elements is equal to the algebraic sum of emf's in that circuit.

i.e. $\boxed{\sum IR = \sum E}$

Consider a circuit is as shown in fig

→ Let us assume current I_1 is flowing in loop abcda & I_2 is flowing in loop cefdc.



→ For the closed loops in the circuit using KVL, the equations are.

For loop abcda: $-I_1 R_1 - (I_1 - I_2) R_2 + E_1 = 0 \rightarrow \textcircled{1}$

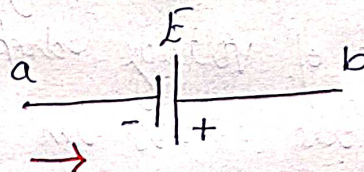
For loop cefdc: $-I_2 R_3 - E_2 - (I_2 - I_1) R_2 = 0 \rightarrow \textcircled{2}$

∴ Equation number $\textcircled{1}$ & $\textcircled{2}$ are called loop equations for the circuit.

→ solve equation $\textcircled{1}$ & $\textcircled{2}$, we get the loop current I_1 & I_2 .

* Signs of Emf's & Voltage drops:

* Signs of Emf's:



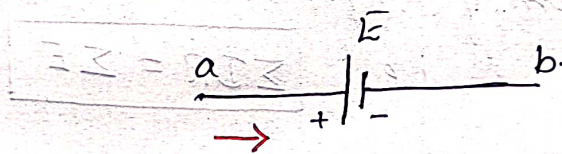
→ If we face the negative terminal of a battery (or) source, while tracing a closed path for KVL, that voltage (or) emf is taken as +ve.

∴ E_{ab} is +ve i.e. $E_{ab} = +E$

→ If we face the positive terminal of a battery (or) source, while tracing a closed path for KVL, that voltage (or) emf is taken as -ve.

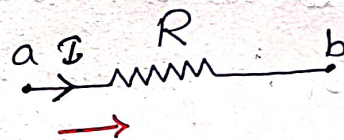
i.e. E_{ab} is -ve.

∴ $E_{ab} = -E$



* Signs of voltage Drops:

→ Consider a Resistance R connected b/w two points a & b , through which a current I is flowing as shown in fig.



→ The voltage drop $(IR)_{ab}$ is along the direction of current, that voltage drop must be taken as -ve.

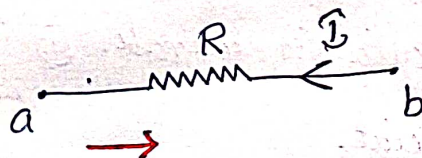
i.e. $(IR)_{ab}$ is -ve

∴ $V_{ab} = -(IR)_{ab}$

→ The voltage drop $(IR)_{ab}$ is against the direction of current, that voltage drop must be taken as +ve.

i.e. $(IR)_{ab}$ is +ve ∴ V_{ab}

∴ $V_{ab} = +(IR)_{ab}$



Problems: [Loop current method]

17 For the fig calculate the current in 2Ω Resistor.

Soln: Apply KVL for loop BCFAB

$$-3I_1 - 2(I_1 - I_2) + 35 = 0$$

$$-5I_1 + 2I_2 = -35 \longrightarrow \textcircled{1}$$

Apply KVL for loop CDEFC

$$-4I_2 - 40 - 2(I_2 - I_1) = 0$$

$$2I_1 - 6I_2 = 40 \longrightarrow \textcircled{2}$$

solve eqn $\textcircled{1}$ & $\textcircled{2}$, we get I_1 & I_2

$$-5I_1 + 2I_2 = -35 \times 2 \Rightarrow -10I_1 + 4I_2 = -70$$

$$2I_1 - 6I_2 = 40 \times -5 \Rightarrow \begin{matrix} (+) & (-) & (+) \end{matrix} \underline{-10I_1 + 30I_2 = -200}$$

$$-26I_2 = 130$$

$$\boxed{I_2 = -5A}$$

The -ve sign for the current I_2 means that it is flowing in the direction opposite to that assumed. i.e it actually flows from D to C in circuit.

• substitute value I_2 in eqn $\textcircled{1}$ we get

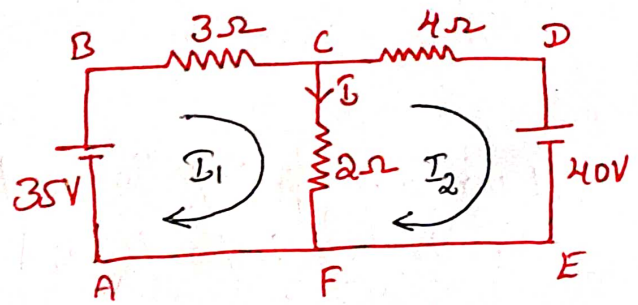
$$-5I_1 + 2(-5) = -35 \Rightarrow -5I_1 = -35 + 10 = -25 \Rightarrow I_1 = \frac{-25}{-5}$$

$$\therefore \boxed{I_1 = 5A}$$

$\therefore$ current flowing through resistor 2Ω is I .

$$\therefore I = I_1 - I_2 = 5 - (-5) = 5 + 5 = 10A$$

$$\boxed{I_{2\Omega} = 10A}$$



2) In the network shown in fig determine all branch currents & voltage across the 5Ω Resistor.

Soln:

Let us assume loop current in the two loops to be I_1 & I_2

→ Apply KVL for loop ABEFA

$$-3I_1 - 5(I_1 - I_2) - 6I_1 + 50 = 0$$

$$-14I_1 + 5I_2 = -50 \rightarrow (1)$$

→ Apply KVL for loop BCDEB

$$-2I_2 - 25 - 8I_2 - 5(I_2 - I_1) = 0$$

$$5I_1 - 15I_2 = 25 \rightarrow (2)$$

Solve eqn (1) & (2), we get I_1 & I_2 .

$$-14I_1 + 5I_2 = -50 \times 5 \Rightarrow -70I_1 + 25I_2 = -250$$

$$5I_1 - 15I_2 = 25 \times -14 \Rightarrow \begin{matrix} (-) & (+) & (-) & (+) \end{matrix} \quad \underline{-70I_1 + 210I_2 = -350}$$

$$-185I_2 = 100 \Rightarrow I_2 = \frac{100}{-185}$$

$$\boxed{I_2 = -0.54A}$$

Substitute I_2 in eqn (1), we get I_1

$$-14I_1 + 5(-0.54) = -50 \Rightarrow -14I_1 = -50 + 2.7 = -47.3$$

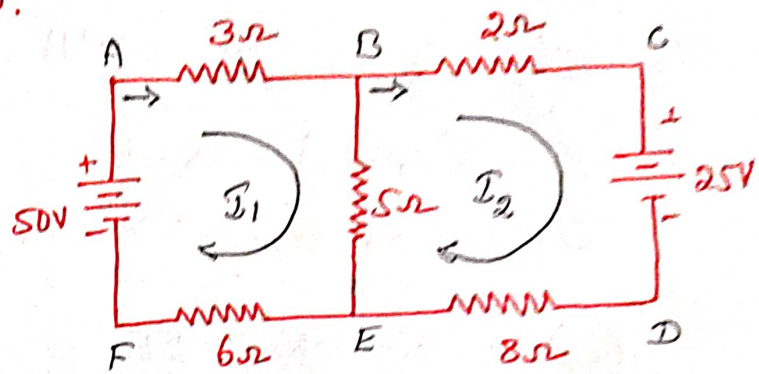
$$\therefore I_1 = \frac{-47.3}{-14} = 3.378A$$

$$\therefore \boxed{I_1 = 3.378A}$$

→ current through 5Ω Resistor, $I = I_1 - I_2 = 3.378 - (-0.54)$

$$\boxed{I_{5\Omega} = 3.918A}$$

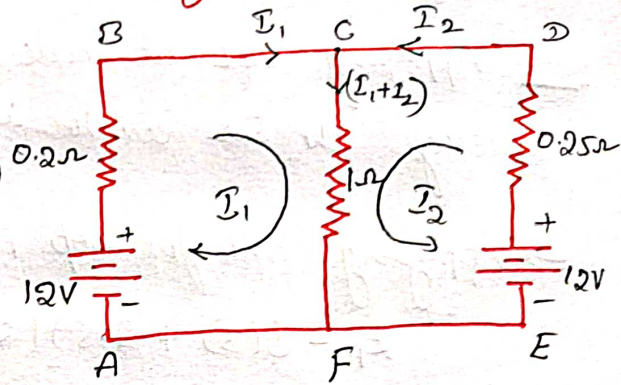
→ voltage drop across 5Ω Resistor $= 5 \times I_{5\Omega} = 5 \times 3.918 = 19.59V$



3) Two 12V batteries with internal resistance 0.2Ω & 0.25Ω respectively are joined in parallel & a resistance of 1Ω is placed across the terminals. find current supplied by each battery.

Soln: Let I_1 & I_2 are the current supplied by the batteries

→ $\therefore$ current through 1Ω Resistor is obtained
by apply KCL at node C.
 $= (I_1 + I_2)$



→ Apply KVL to loop BCFA B.

$$- (I_1 + I_2)1 + 12 - I_1(0.2) = 0$$

$$-1.2I_1 - I_2 = -12 \rightarrow \textcircled{1}$$

→ Apply KVL to loop CDEFC.

$$- (I_1 + I_2)1 + 12 - I_2(0.25) = 0$$

$$-I_1 - I_2 \cdot 1.25 = -12 \rightarrow \textcircled{2}$$

Solve eqn $\textcircled{1}$ & $\textcircled{2}$

$$-1.2I_1 - I_2 = -12 \times -1 \Rightarrow 1.2I_1 + I_2 = 12$$

$$-I_1 - 1.25I_2 = -12 \times -1.2 \Rightarrow 1.2I_1 + 1.5I_2 = 12 \cdot 1.2$$

$$-0.5I_2 = -2.4$$

$$\boxed{I_2 = 4.8 \text{ A}}$$

Substituting the values of I_2 in eqn $\textcircled{1}$, we get

$$-1.2I_1 - 4.8 = 12 \Rightarrow 1.2I_1 = 7.2 \Rightarrow \boxed{I_1 = 6 \text{ A}}$$

* Problems (Branch current method)

4) In the circuit shown in fig find E_1 , E_2 & I when the power dissipated in the 5Ω resistor is $125W$.

Soln: Applying KCL at node A, the current through the 5Ω Resistor is $= (I + 2.5)$

→ Applying KVL for loop ABCFA

$$E_1 - I(3) + (2.5)(2) - E_2 = 0$$

$$E_1 - E_2 = 3I - 5 \rightarrow (1)$$

→ Applying KVL for loop FCDEF

$$E_2 - (2.5)(2) - (I + 2.5)5 = 0$$

$$E_2 = 5I + 17.5 \rightarrow (2)$$

given that power dissipated in 5Ω Resistor is $125W$

$$\text{i.e. } P = 125W$$

current through 5Ω Resistor is $= (I + 2.5)$

$$\therefore I_{5\Omega}^2 R = 125$$

$$(I + 2.5)^2 \times 5 = 125 \Rightarrow (I + 2.5)^2 = 25$$

$$(I + 2.5) = 5 \Rightarrow I = 5 - 2.5 = 2.5$$

$$\therefore \boxed{I = 2.5A}$$

substitute value of I in eqn (2) we get

$$E_2 = 5(2.5) + 17.5 = 30V$$

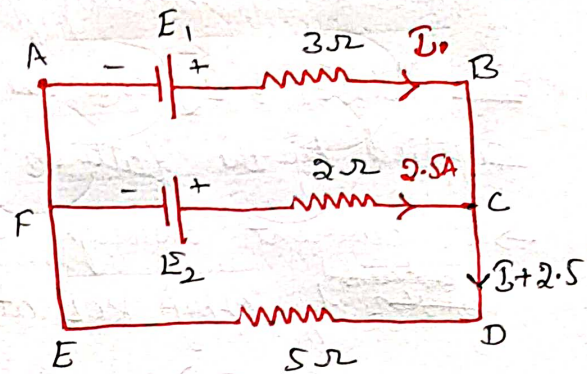
$$\therefore \boxed{E_2 = 30V}$$

substitute value of I & E_2 in eqn (1) we get

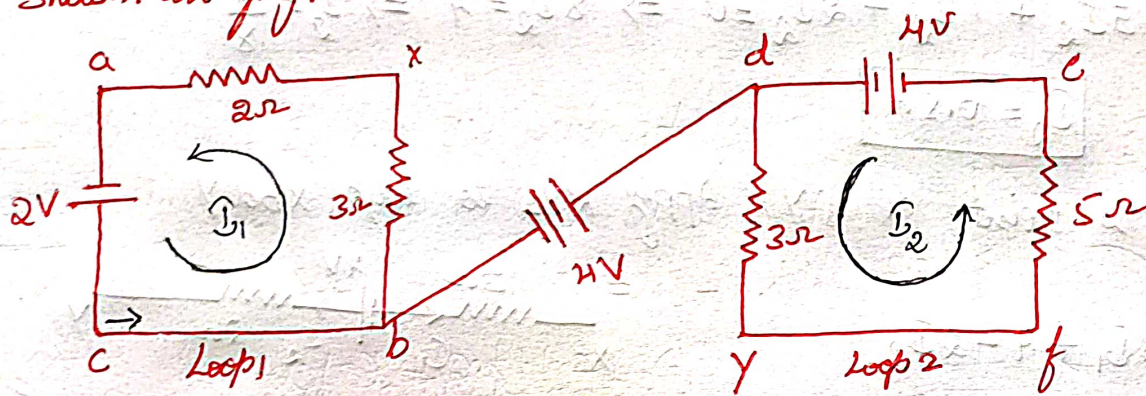
$$E_1 - 30 = (3 \times 2.5) - 5 = 2.5$$

$$E_1 = 2.5 + 30 = 32.5V$$

$$\boxed{E_1 = 32.5V}$$



5) Find the potential difference b/w the point x & y in the network shown in fig.



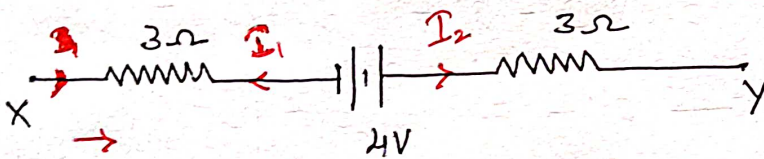
Assume current I_1 in loop 1 & I_2 in loop 2

→ Apply KVL for loop a-x-b-c-a

$$-3I_1 - 2I_1 + 2 = 0 \Rightarrow -5I_1 = -2 \Rightarrow I_1 = \frac{2}{5} \Rightarrow \boxed{I_1 = 0.4A}$$

→ Apply KVL for loop d-e-f-y-d

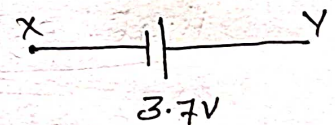
$$-3I_2 - 5I_2 + 4 = 0 \Rightarrow -8I_2 = -4 \Rightarrow I_2 = \frac{4}{8} \Rightarrow \boxed{I_2 = 0.5A}$$



$$V_{xy} = +3I_1 + 4 - 3I_2 = 3(0.4) + 4 - 3(0.5)$$

$$\boxed{V_{xy} = 3.7V}$$

y is at higher potential than x.



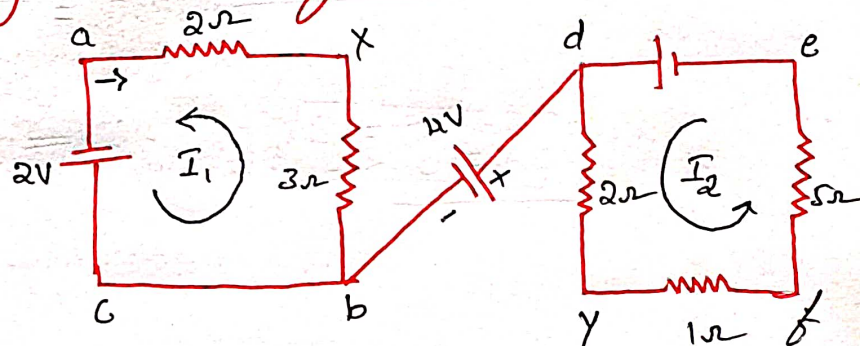
6) For the circuit shown in fig obtain voltage between point x & y.

Soln: Assume current I_1 & I_2 in loop 1 & loop 2.

Apply KVL to loop-1: a-x-b-c-a

$$2I_1 + 3I_1 - 2 = 0$$

$$5I_1 = 2 \Rightarrow I_1 = \frac{2}{5} = 0.4A, \quad \boxed{I_1 = 0.4A}$$



→ Apply KVL to loop 2: def'd.

$$-4 + 5I_2 + 1I_2 + 2I_2 = 0 \Rightarrow 8I_2 = 4 \Rightarrow I_2 = \frac{4}{8}$$

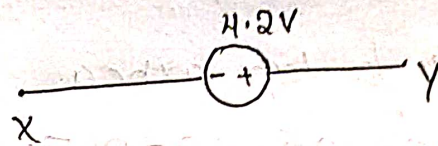
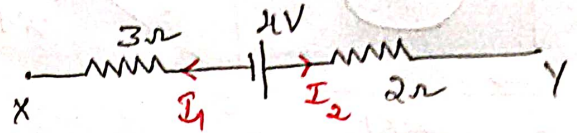
$$I_2 = 0.5A$$

To find V_y b/w point x & y, Apply KVL to loop xbdy

$$\therefore V_{xy} = 3I_1 + 4 - 2I_2$$

$$V_{xy} = 3(0.4) + 4 - 2(0.5)$$

$$V_{xy} = 4.2V$$

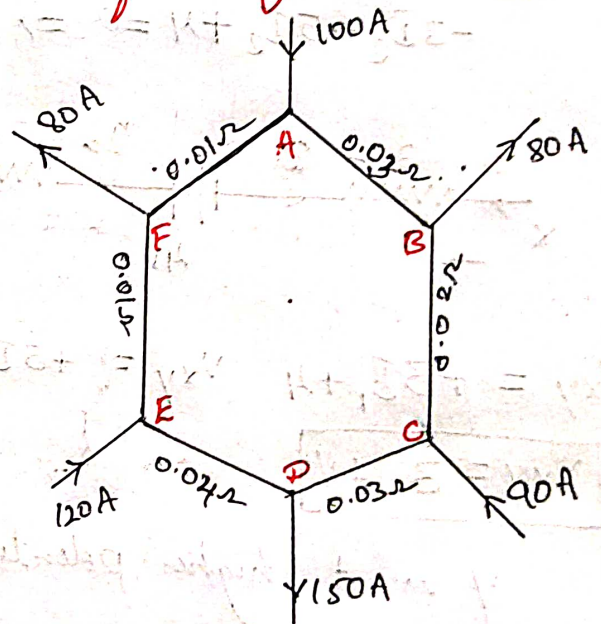
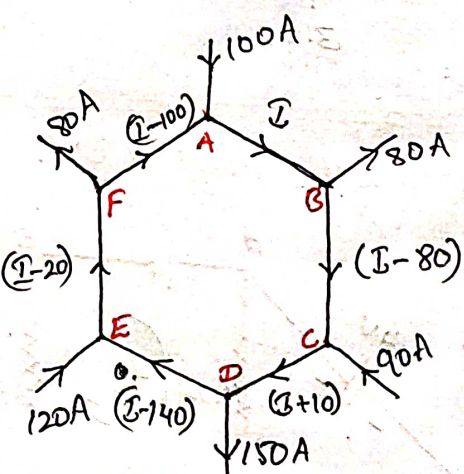


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7) Find the currents in circuits branches of the given network shown in fig.

Soln: let current in branch AB be I , as shown in fig below.

Apply KCL at node B, C, D, E & F the branch currents are shown below



* POWER & ENERGY:

* Electrical Work:

→ In an electrical circuit, there is movement of electrons which constitutes flow of current. This movement of electrons results in transfer of charge.

→ Electrical work is done when there is a transfer of charge.

→ The unit of Electrical work done is "Joule"

→ One joule of electrical work done is that "work done in moving a charge of 1 Coulomb through a potential difference of 1 Volt."

$$\therefore \boxed{\text{Electrical work, } W = V \times Q} \text{ Joules}$$

where $V \rightarrow$ potential difference in Volts.

$Q \rightarrow$ charge in Coulombs.

$$\text{Now, } I = \frac{Q}{t} \Rightarrow Q = It$$

$$\therefore \boxed{W = VIt} \text{ Joules}$$

where $t \rightarrow$ time in seconds.

* Electrical Power:

→ The rate at which electrical work is done in an electric circuit is called electrical power.

$$\therefore \text{Electrical power, } P = \frac{\text{Electrical work done}}{\text{Time}}$$

$$P = \frac{W}{t} = \frac{VIt}{t}$$

$$\therefore \boxed{P = VI} \text{ Joules/sec @ Watts.}$$

→ The unit of Power is "watt" is very small as compared with the power transfers in practical circuits.

→ Hence Power is generally expressed in bigger unit KW @ MW.

$$1 \text{ KW} = 1000 \text{ watt} \quad \& \quad 1 \text{ MW} = 1 \times 10^6 \text{ watt}.$$

→ The power consumed in an electrical circuit is one watt, if potential difference applied across the circuit is 1 volt which causes 1 ampere of current to flow through the circuit.

$$\therefore P = VI \text{ watts}$$

$$1 \text{ watt} = 1 \text{ volt} \times 1 \text{ ampere}.$$

$$1 \text{ watt} = 1 \text{ Joules/sec}.$$

* Electrical Energy:

→ Electrical Energy is the total amount of electrical work done in an electrical circuit.

$$\therefore \text{Electrical Energy} = \text{Power} \times \text{time}.$$

$$\text{i.e. } W = VI t \text{ Joules @ watt-sec}.$$

→ The unit of Electrical Energy is "Joules" @ "watt-sec"

→ Energy consumed by an electric circuit is 1 Joules, when it utilises power of 1 watt for 1 sec.

→ The watt-sec is very small unit, electrical Energy is measured in larger units, i.e. watt-hour & "kilowatt-hour (kWh)"

$$\rightarrow 1 \text{ unit} = 1 \text{ kWh} = 1 \times 10^3 \times 3600$$

$$\rightarrow \therefore 1 \text{ kWh} = 3.6 \times 10^6 \text{ Joules}.$$

Power & Energy using Ohm's Law:

→ Electrical Power is given by $P = VI \rightarrow (1)$

According to Ohm's Law, $V = IR$. & $I = \frac{V}{R}$.

Substituting in eqn (1) we get.

$$\boxed{P = I^2 R} \quad \text{or} \quad \boxed{P = \frac{V^2}{R}} \quad \text{Watts}$$

→ Electrical Energy is $W = VIt \rightarrow (2)$

Substituting V & I in eqn (2), we get.

$$\boxed{W = I^2 R t} \quad \text{or} \quad \boxed{W = \frac{V^2}{R} t} \quad \text{Joules.}$$

The above expressions are used to calculate the power consumed in the various parts of an electric circuit.

Problems:

1) A 250V lamp is rated to pass a current of 0.30A. find its power output. ^{now} if a second similar lamp is connected in parallel with first lamp. (i) find the supply current required to give the same power output in each lamp. (ii) Also calculate power, if lamps are connected in series.

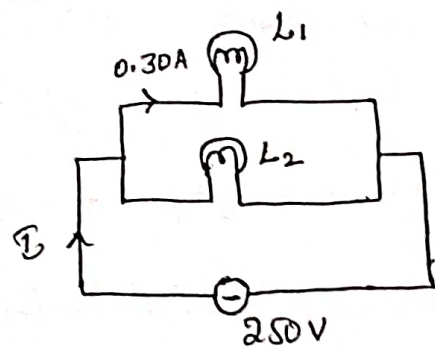
Soln: (i) When lamps are in parallel,
Power output of lamp is

$$P_1 = VI = 250 \times 0.30 = 75W.$$

When the 2nd 11th lamp is connected in parallel.

→ Total power $P_t = P_1 + P_2 = 75 + 75 = 150W$.

→ Supply current $I = \frac{P_t}{V} = \frac{150}{250} = 0.60A$.

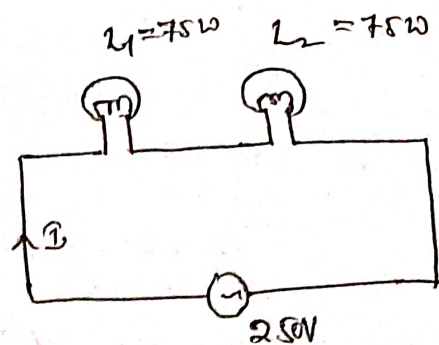


ii) When Lamps are Connected in Series

In case of one Lamp.

$$\text{Resistance of Lamp } R = \frac{V}{I} = \frac{250}{0.30}$$

$$R = 833.3 \Omega$$



When both the lamps are Connected in Series, the total Resistance is

$$R_t = 833.3 + 833.3 = 1666.60 \Omega$$

$$I = \frac{V}{R_t} = \frac{250}{1666.60} = 0.15A$$

$$\therefore P_T = VI = 250 \times 0.15 = 37.5W$$

(OR)

Total Power

$$P_T = \frac{P_1 \times P_2}{P_1 + P_2}$$

$$P_T = \frac{75 \times 75}{75 + 75} = 37.5W$$

* Note * When Lamps P_1 & P_2 are Connected in Series, the

$$\text{total power is } P_T = \frac{P_1 P_2}{P_1 + P_2}$$

* When Lamps P_1 & P_2 are Connected in Parallel, the total power

$$\text{is } P_T = P_1 + P_2$$

2) A current of 4A passes through a 10Ω Resistor, calculate

(a) the power developed by the resistor

(b) The energy dissipated in 3 minutes.

Soln (a) $P = I^2 R = (4)^2 \times 10 = 160W$

(b) $W = I^2 R t = (4)^2 \times 10 \times 3 = 2880J$

3) A heater is given a current of 10A from a 250V source for 15 hours, Determine the energy consumed by the heater in kWh.

Soln $I = 10A$, $V = 250$, $t = 15hr$.

$$P = VI = 250 \times 10 = 2500W = 2.5kW$$

$$W = Pt = 2.5 \times 10^3 \times 15 = 37.50kWh$$

For the ckt shown in fig determine the power developed by each resistor.

Soln: Apply current division Rule.

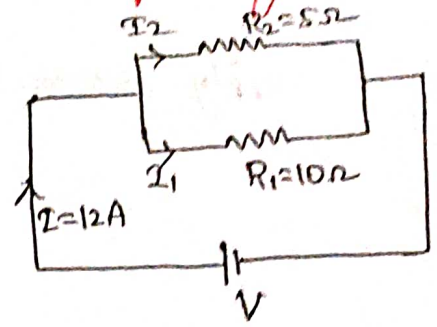
$$I_1 = \frac{I \times R_2}{R_1 + R_2} = \frac{12 \times 5}{10 + 5} = 4 \text{ A}$$

$$\therefore I_2 = I - I_1 = 12 - 4 = 8 \text{ A}$$

hence power developed by Resistor R_1 & R_2 is

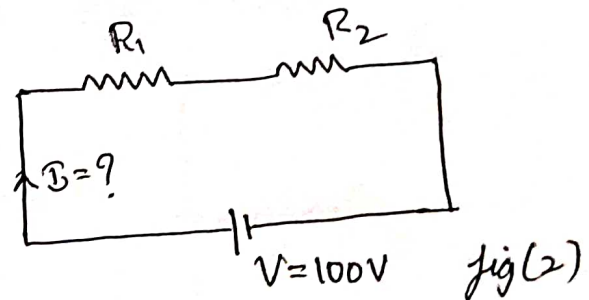
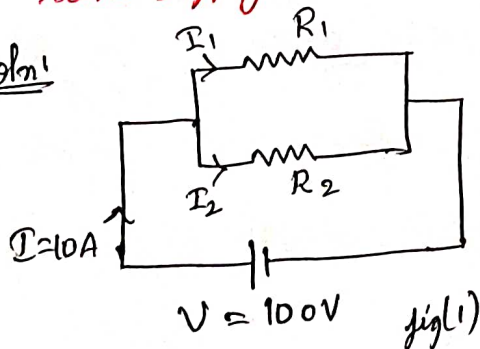
$$P_1 = I_1^2 R_1 = (4)^2 \times 10 = 160 \text{ W}$$

$$P_2 = I_2^2 R_2 = (8)^2 \times 5 = 320 \text{ W}$$



5) Two resistors connected in parallel across 100V D.C supply take 10A from the line. The power dissipated in one resistor is 600W. What is the current drawn when they are connected in series across the same supply?

Soln:



Power dissipated in Resistor R_1 i.e $P_1 = 600 \text{ W}$.

$$\therefore \text{WKT } P_1 = V_1 I_1 = 100 \times I_1$$

$$\therefore I_1 = \frac{P_1}{V_1} = \frac{600}{100} = 6 \text{ A}$$

$$I_2 = I - I_1 = 10 - 6 = 4 \text{ A}$$

$$\therefore R_1 = \frac{V}{I_1} = \frac{100}{6} = 16.66 \Omega$$

$$R_2 = \frac{V}{I_2} = \frac{100}{4} = 25 \Omega$$

Now R_1 & R_2 are connected in series in fig(2) across the 100V D.C supply.

$\therefore V = V_1$ is
Parallel ckt

$$\text{Total Resistance } (R_T) = R_1 + R_2 = 16.66 + 25 = 41.66 \Omega$$

$$\therefore \text{Supply current } (I) = \frac{V}{R_T} = \frac{100}{41.66} = 2.4 \text{ A}$$

$$\boxed{I = 2.4 \text{ A}}$$



Power & Energy Equations:

* Electric Power (P): $P = \frac{\text{Electrical work done}}{\text{time}} = \frac{VIt}{t} = VI$

$$\boxed{P = VI} \quad \text{or} \quad \boxed{P = I^2 R} \quad \text{or} \quad \boxed{P = \frac{V^2}{R}}$$

* Electrical Energy (W): $W = \text{Power} \times \text{time}.$

$$\therefore \boxed{W = VIt} \quad \text{or} \quad \boxed{W = I^2 R t} \quad \text{or} \quad \boxed{W = \frac{V^2}{R} t}$$

* Unit of Electric power is joules/sec i.e. "Watt"

* Unit of Electrical Energy is "kWh" (kilowatt-hour).

Formulae:

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→ Ohm's Law: a) $V = IR$

b) $I = \frac{V}{R}$

c) $R = \frac{V}{I}$



→ Series circuit: $V = V_1 + V_2 + V_3 \dots$

$I = I_1 = I_2 = I_3$ (current is same in all elements which are connected in series).

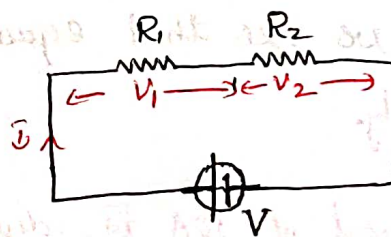
→ Parallel circuit: $I = I_1 + I_2 + I_3 \dots$ is same

$V = V_1 = V_2 = V_3$ (voltage across each elements which are connected in parallel).

→ voltage division Rule: Applicable for series circuit

* voltage across resistor R_1 (V_1)

$$= \frac{\text{Main voltage} \times \text{Resistance of the same Branch}}{\text{Sum of the Resistance}}$$



$$V_1 = \frac{V \times R_1}{R_1 + R_2}$$

||

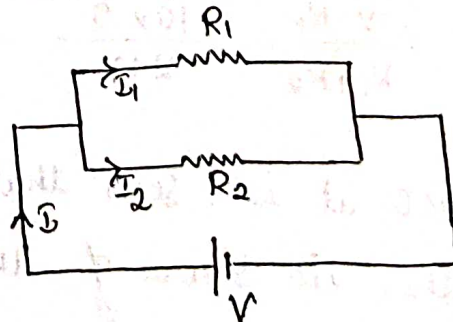
$$V_2 = \frac{V \times R_2}{R_1 + R_2}$$

→ current division Rule: Applicable for parallel circuit.

$$I = I_1 + I_2$$

* current through Resistor (R_1) [I_1]

$$= \frac{\text{Main current} \times \text{Resistance of the other Branch}}{\text{Sum of the Resistance.}}$$



$$I_1 = \frac{I \times R_2}{R_1 + R_2}$$

||

$$I_2 = \frac{I \times R_1}{R_1 + R_2}$$